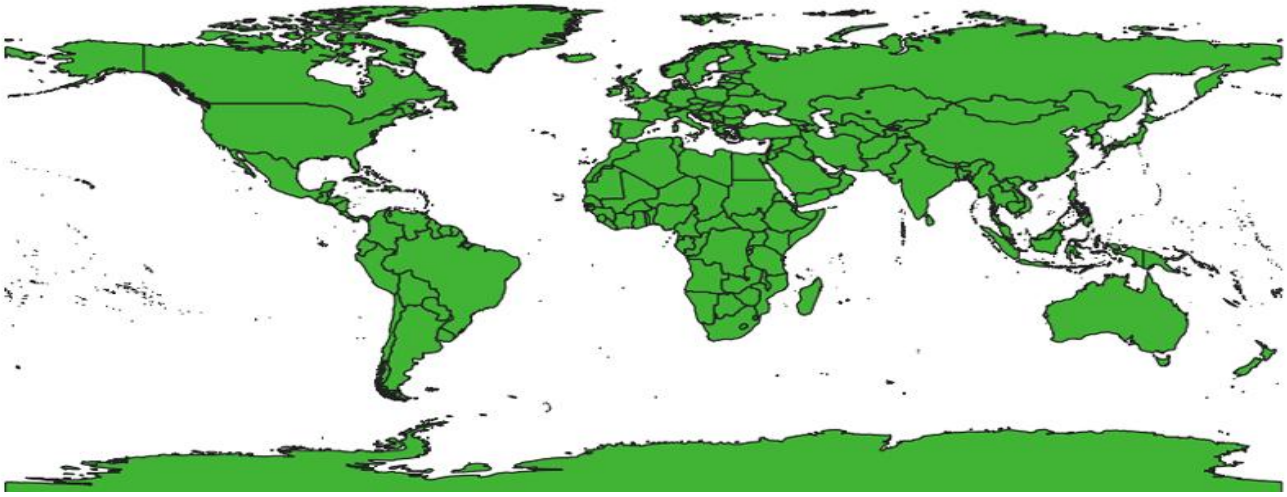
Practical No – 10

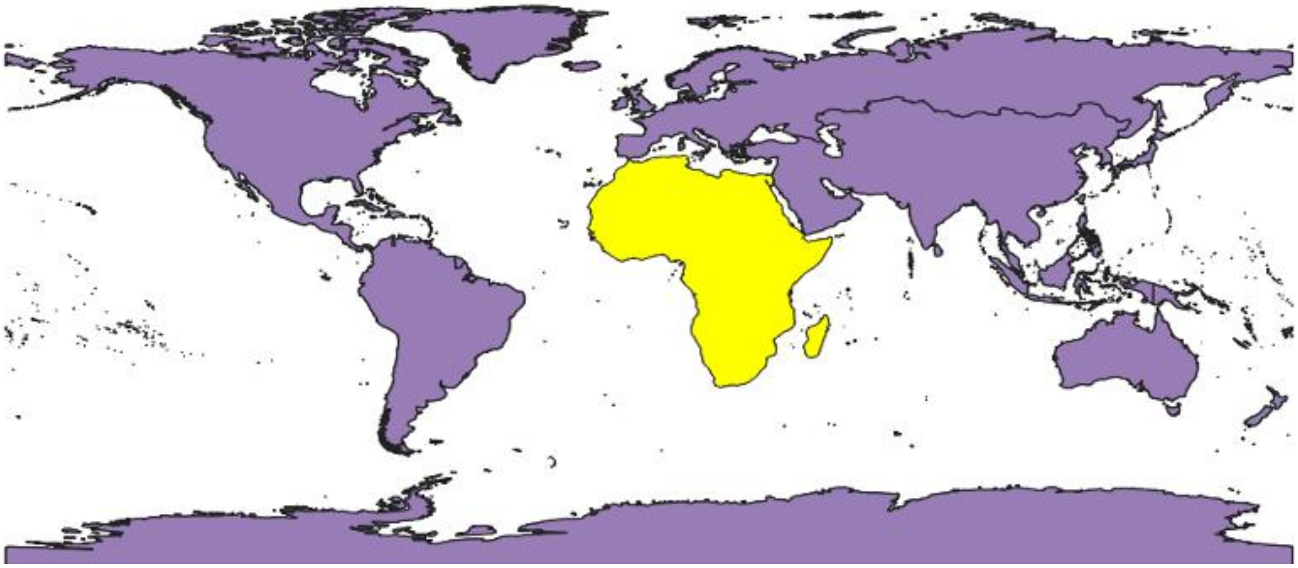
Batch Processing using Processing Framework, Automating Complex Workflows using Processing Modeller & Automating Map Creation with Print Composer Atlas

A. Batch Processing using Processing Framework

- Go to Layer→Add Vector Layer.
- Browse to the downloaded Admin0Countries shape file ne_10m_admin_0_ Countries.shp and click Open.

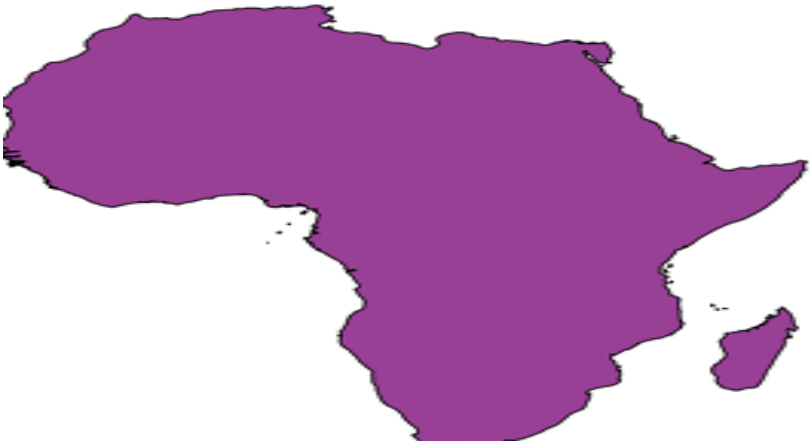


- Vector→Geoprocessing Tools→Dissolve.
- Select ne_10m_admin_0_countries as the Input vector layer. The Dissolve filed would be CONTINENT. Name the output file as continents.shp and check the box next to Add result to canvas.

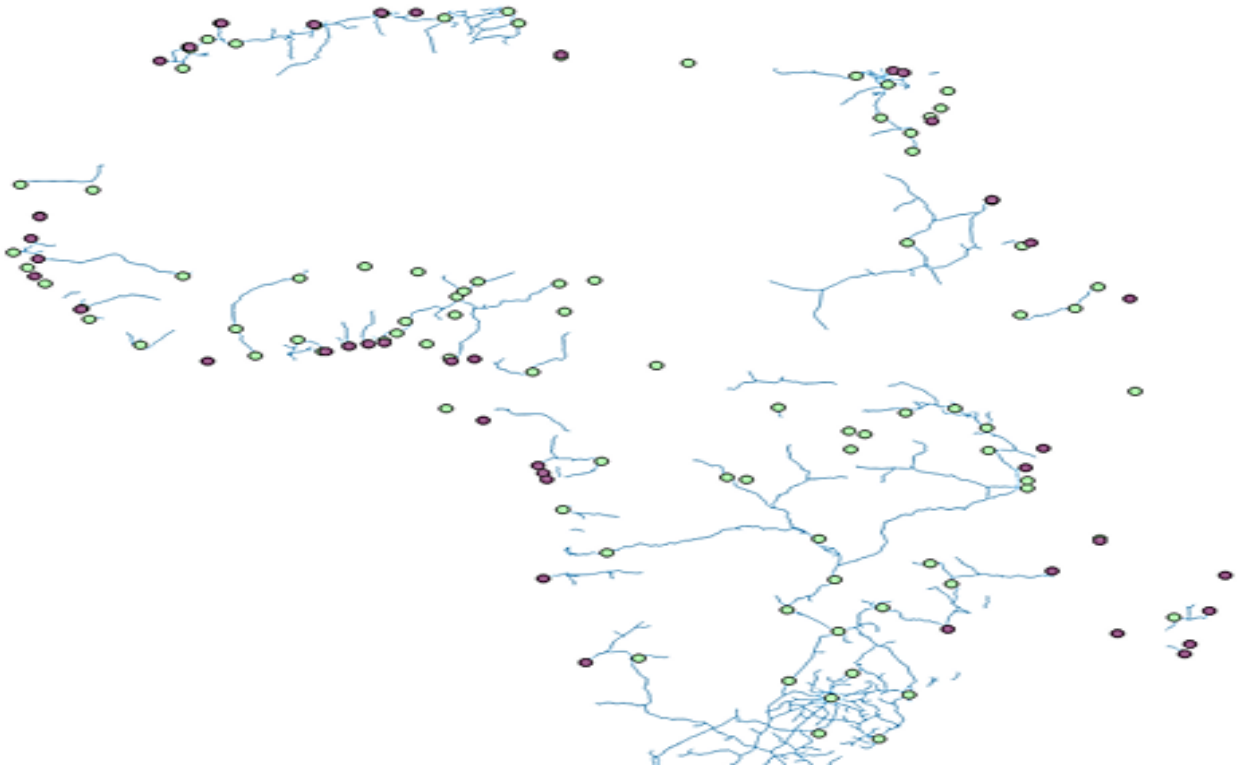


- Right-click the continents layer and select save selection as…Name the output file as africa.shp.

- Start our batch clip process. Open Processing→Tooklbox.

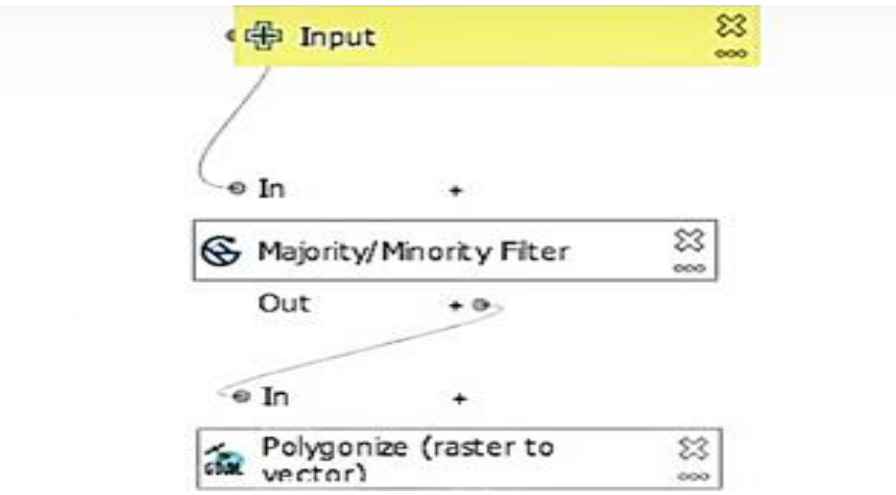


- Vector→Overlay tools→Clip
- Browse to the directory containing the global transportation layer→Select All→Click Open.
- The Input layer columns will be auto-populated with all layer you had selected. You may use add row button to add more rows and define more inputs. Next, we need to select the layer containing the boundary to clip our input layers. Click the…button for the first row and add the africa.shp clip layer.
- Browse the directory where you want output layers. Type the filename as clipped_and click Save.
- Start the batch processing. Click RUN.
- All the global layers are properly clipped to the continent boundary that we had specified.

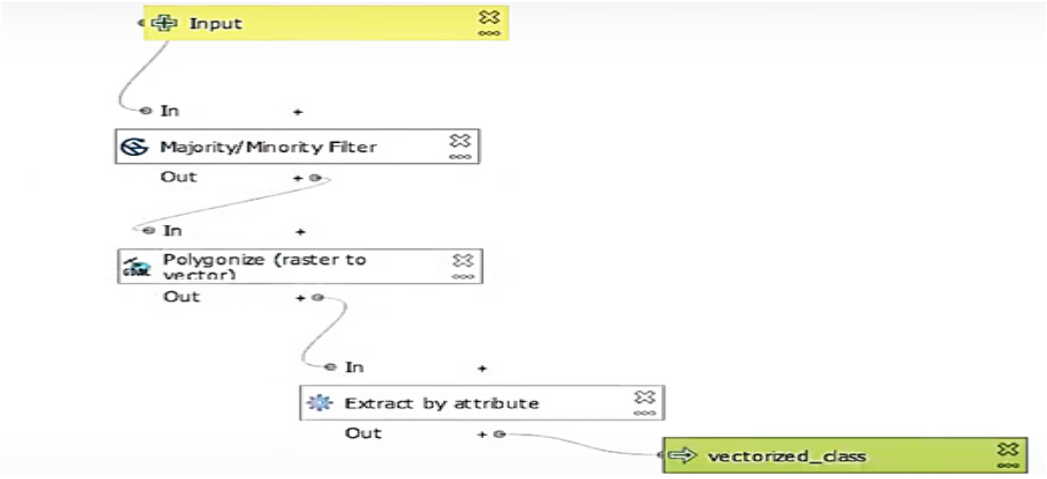


B. Automating Complex Workflows using Processing Modeller

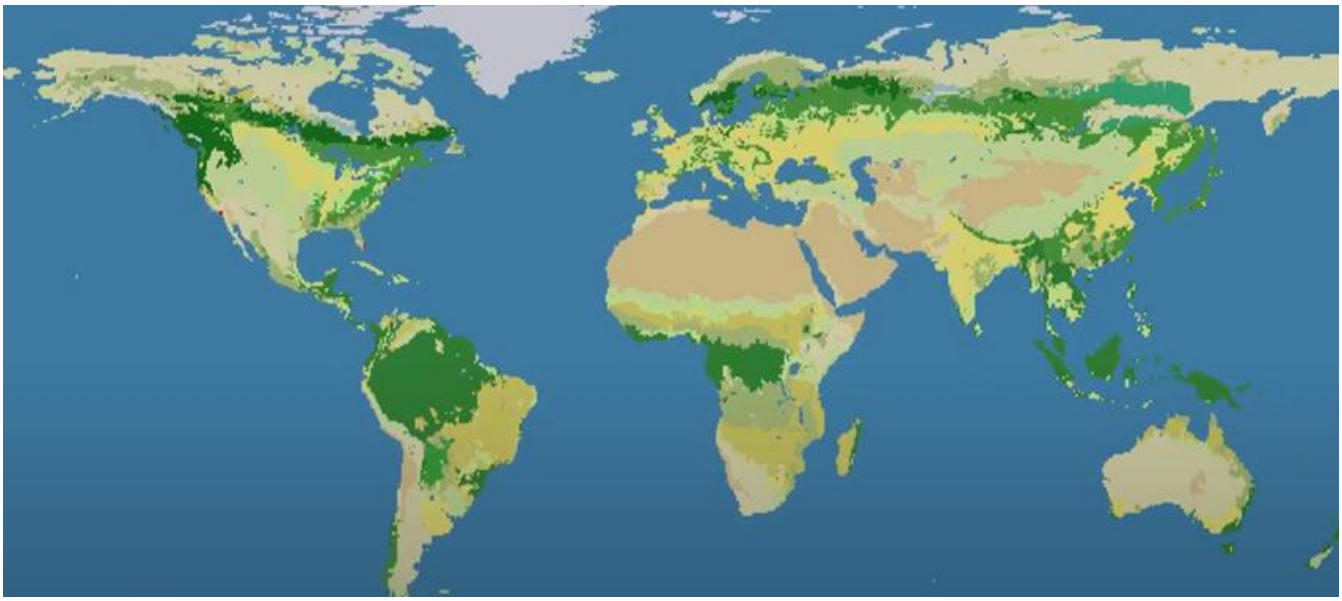
- Go to Processing→Graphical Modeller…
- The Processing modeller dialog contains a left-hand panel and a main canvas. Select the input tab in the left-hand panel and drag the + Raster layer to the canvas
- A Parameter definition dialog will pop-up. Enter Input as the Parameter name and mark Yes to Require. Click OK.
- Next step is to apply a Majority filter algorithm. Switch to the algorithm tab from the bottom-left corner. Search for the algorithm and you will find it listed under SAGA provider. Drag it to the canvas.
- A configuration dialog for Majority Filter will be presented. Leave the values to their default and click OK.
- To convert the output of majority filter to vector. Find the polygonize (raster to vector) algorithm and drag it to the canvas.



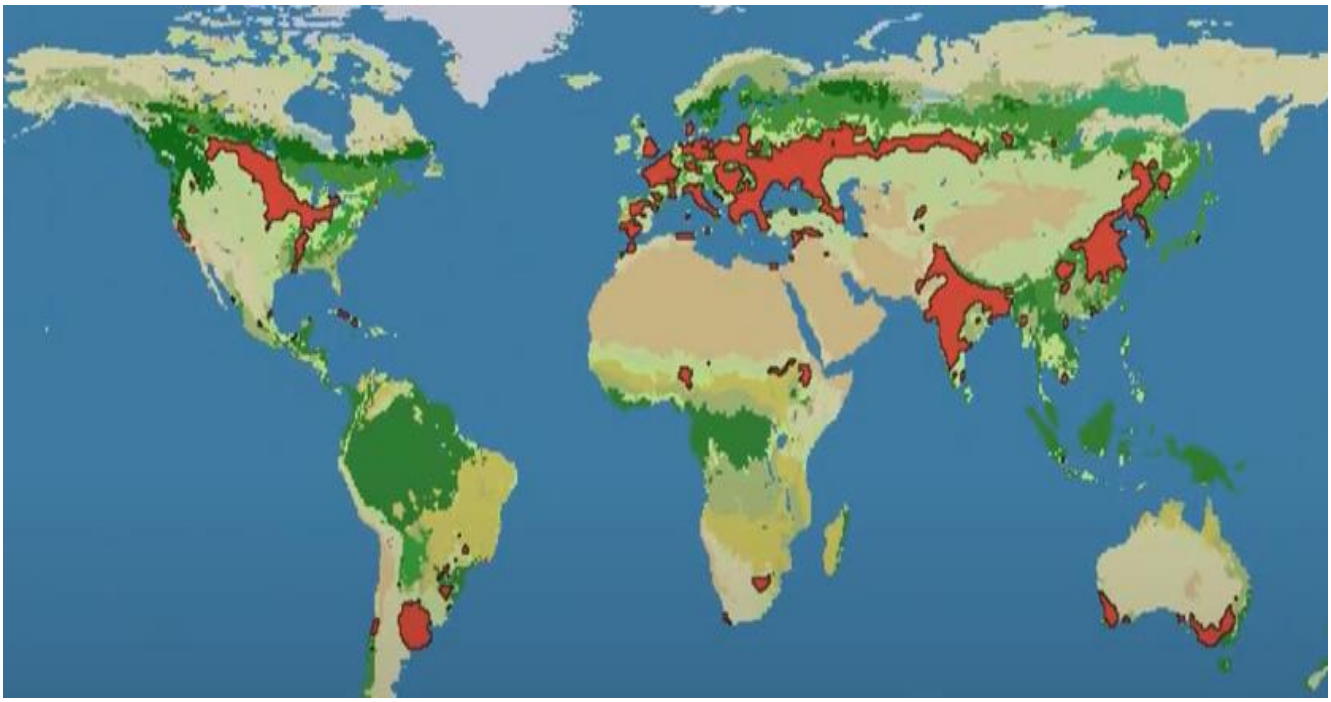
- Select 'Filtered Grid' from algorithm 'Majority Filter' as the value for input layer. Click OK.
- The final step in the workflow is to query for a class value and create a new layer from the matching features. Search for the Extract by attribute algorithm and drag it the canvas.



- Now it is time to rest our model. Close the modeller and switch to the main QGIS window. Go to Layer→Add Layer→Add Raster Layer…
- Browse to the downloaded LC_hd_global_2001.tif.gz file and click Open. Once the raster is loaded, go to processing→Toolbax.

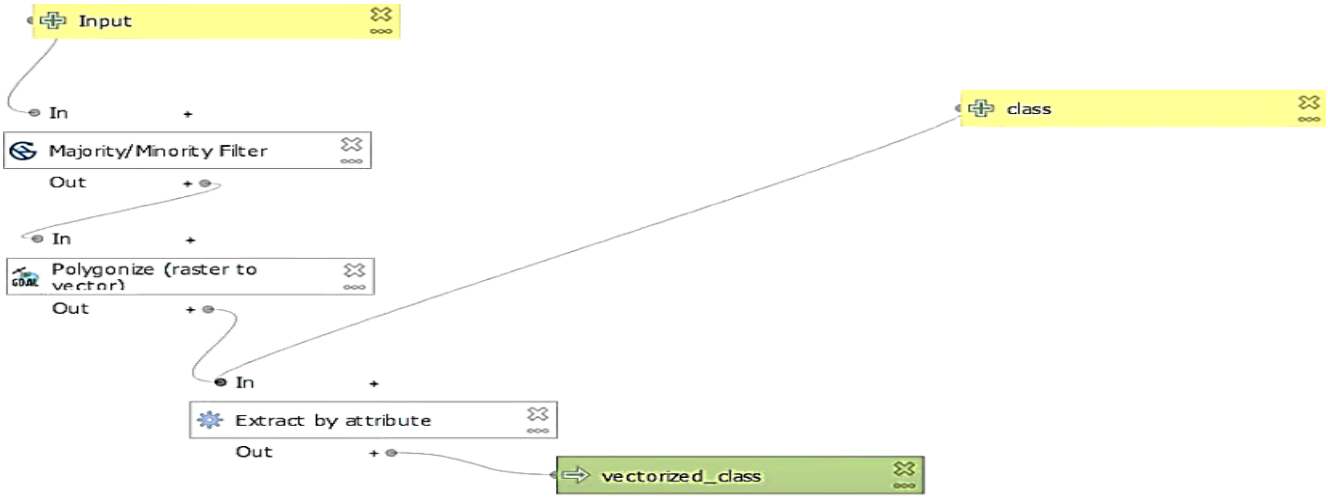


- Find the newly created model under Model→Raster→Vectorize.Double-click to launch the model.
- Select LC_hd_global_2001 as the Input and click Run.
- Right-click on the vectorize model and select Edit model.

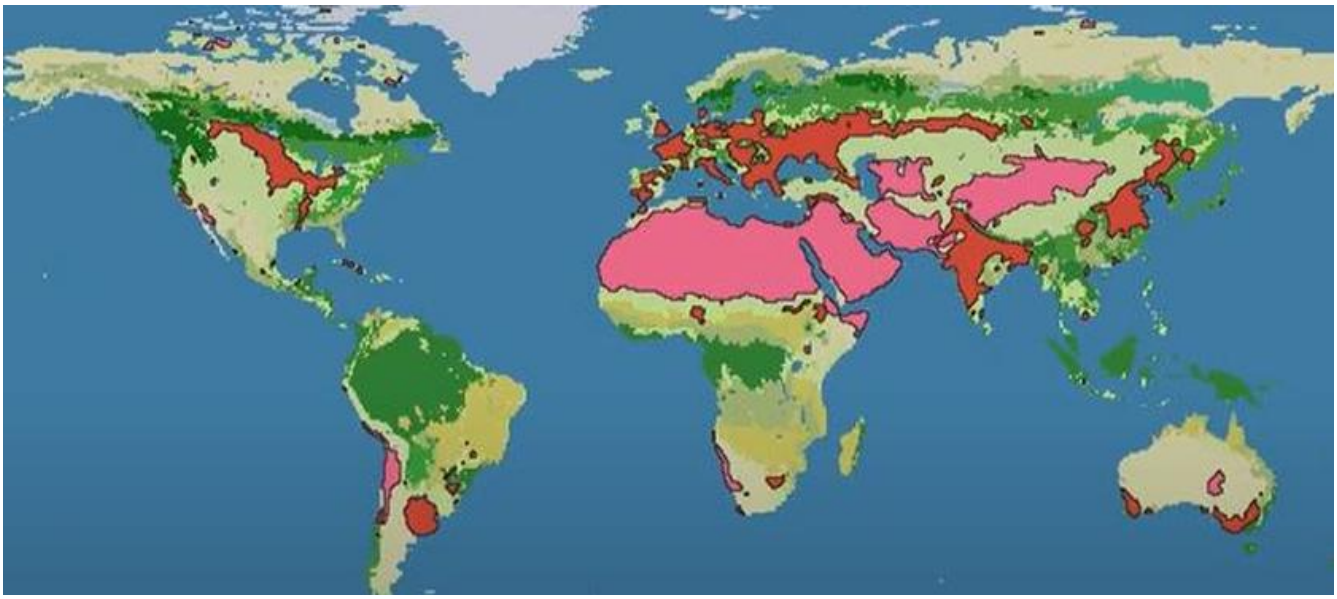


- Enter the Parameter Name as Class. Enter 12 as the Default value.
- We will now change the Extract by attribute algorithm to use this input instead of the hard-coded value. Click the **Edit** button next to the **Extract** by attribute box.

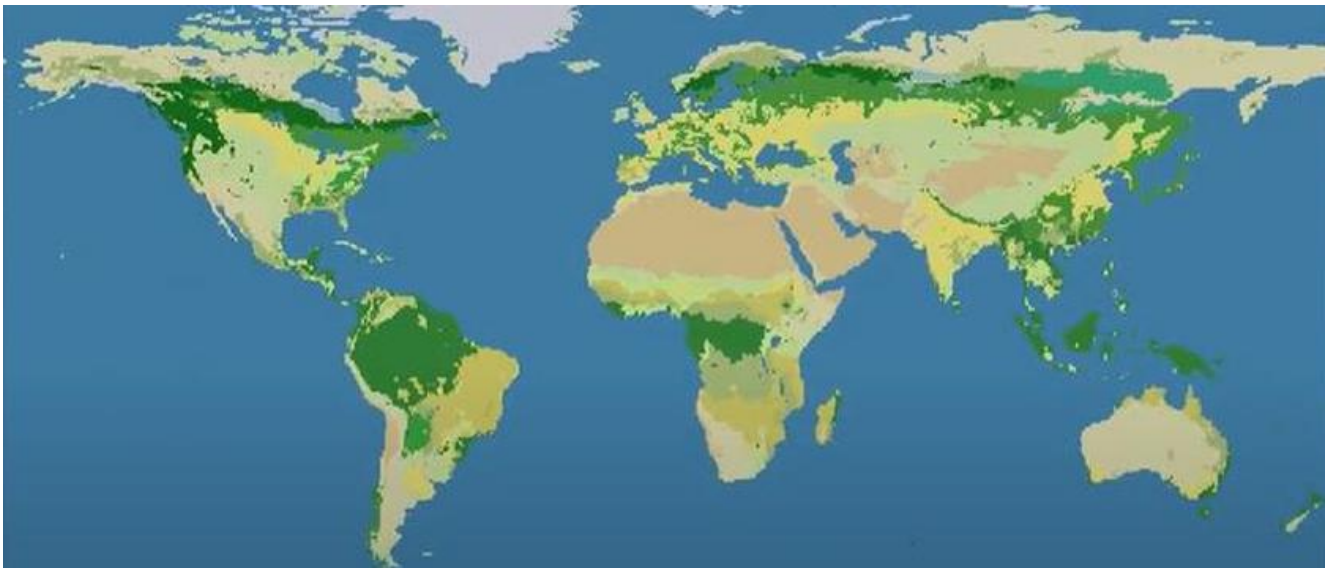
- Click the dropdown arrow for value and select class. Class OK.



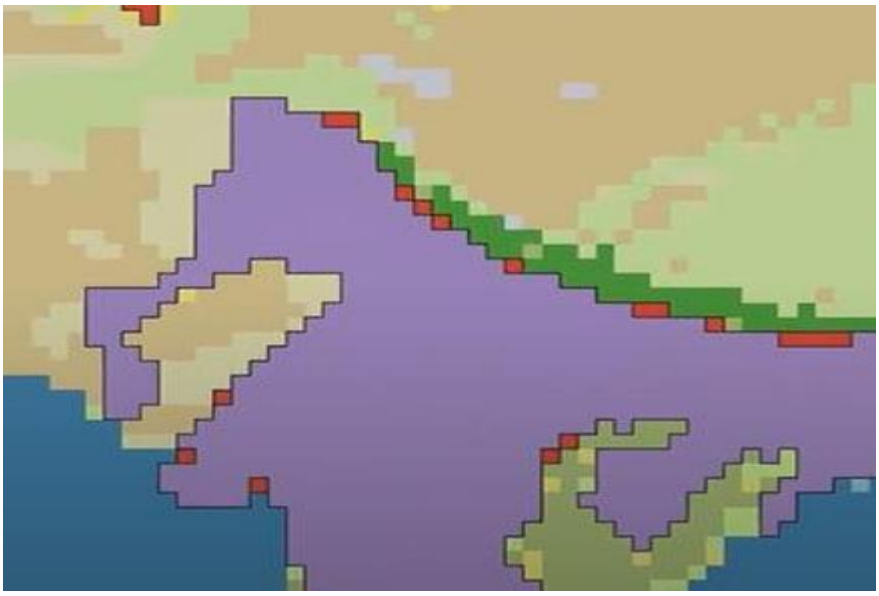
- Enter 16 as the class value and click Run.



- Now model is ready. Load the LC_hd_global_2012.tif.gz file by going to layer→Add Layer→Add Raster Layer...Click the vectorize model from the processing toolbox panel.



- Pick the LC_hd_global_2012 layer as the input and click Run.
- Once the new output is loaded, you can compare the changes in the croplands form 2001 to 2012.

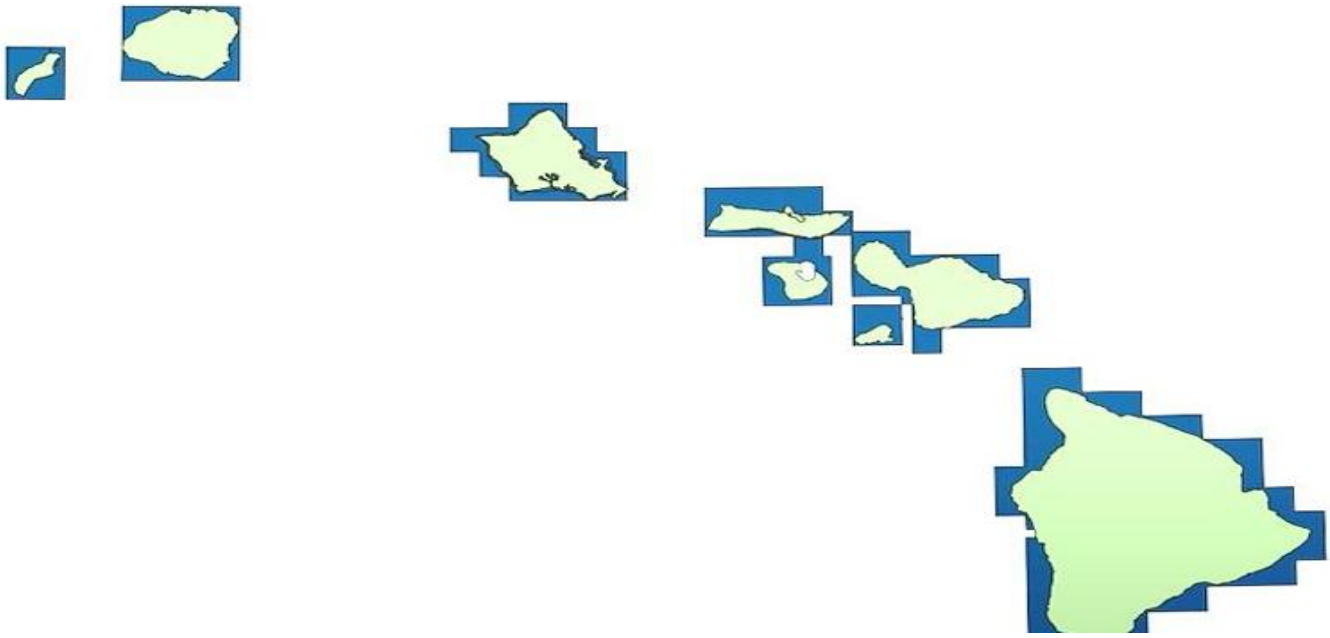


- Right-click the vectorize model and select Edit model. Click the edit model help button from the toolbar.
- In the Help editor dialog, select any item from the select element to edit panel and enter the help text in element description. Click OK. This help will be available in the Help tab when you launch the model to run.

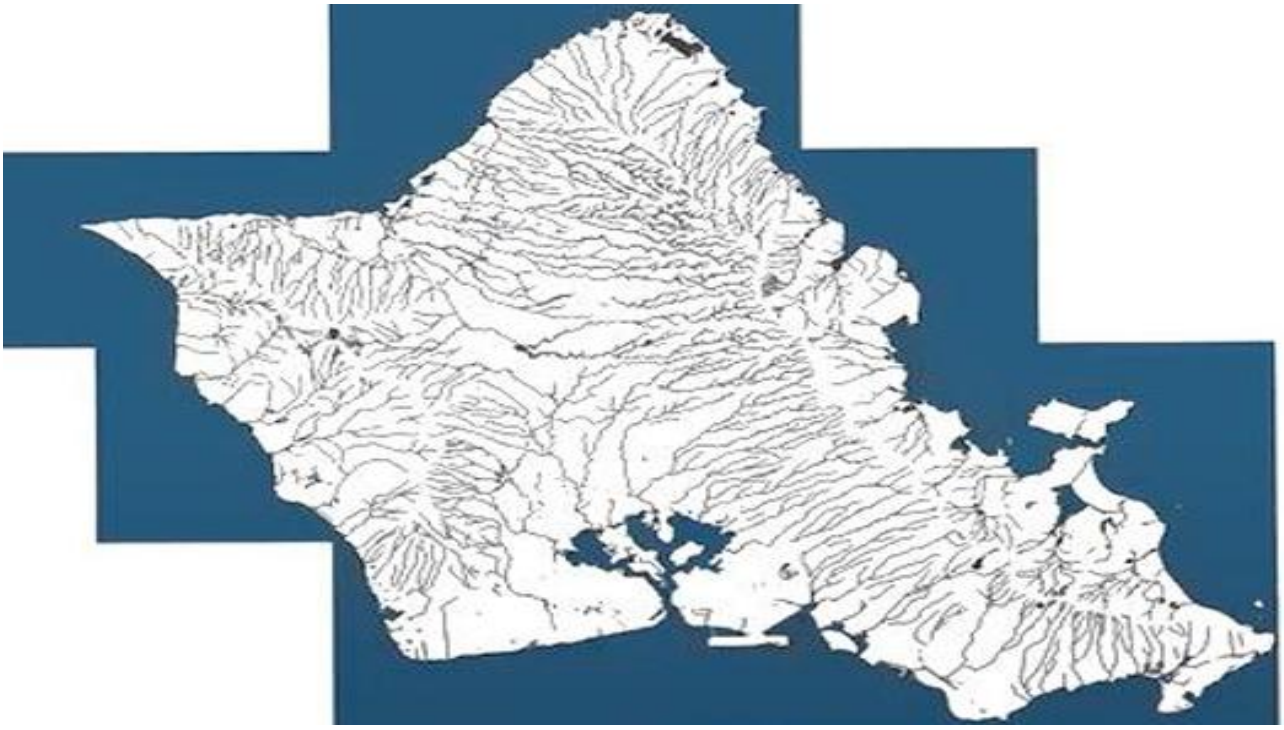
C. Automating Map Creation with Print Composer Atlas

- Launch QGIS and go to Layer→Add Layer→Add Vector Layer. Browse to the HI_Wetlands.shp.zip file and click Open.
- Select the **HI_Wetlands_Poly** layer and click OK.

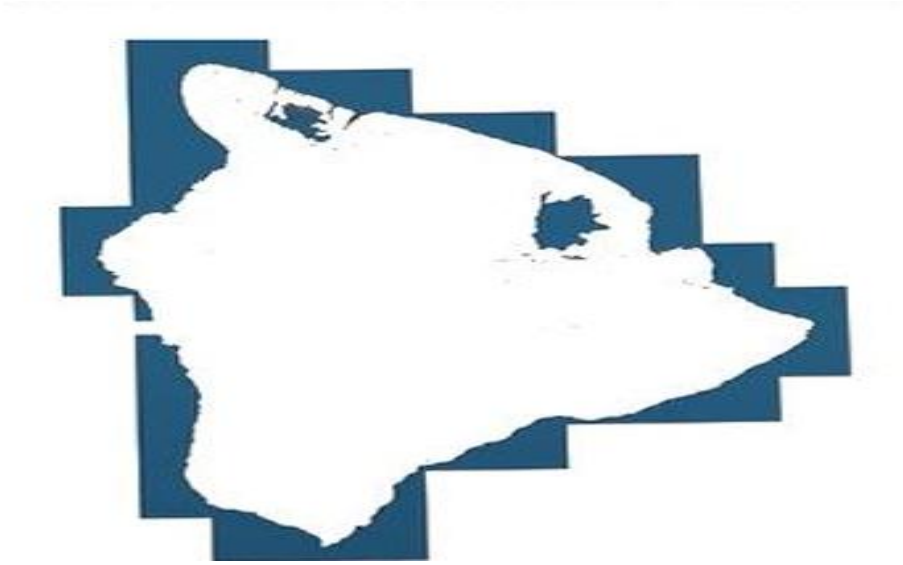
- Go to Layer→Add Layer→Add Vector Layer and browse to the county 10.shp.zip file. Click Open



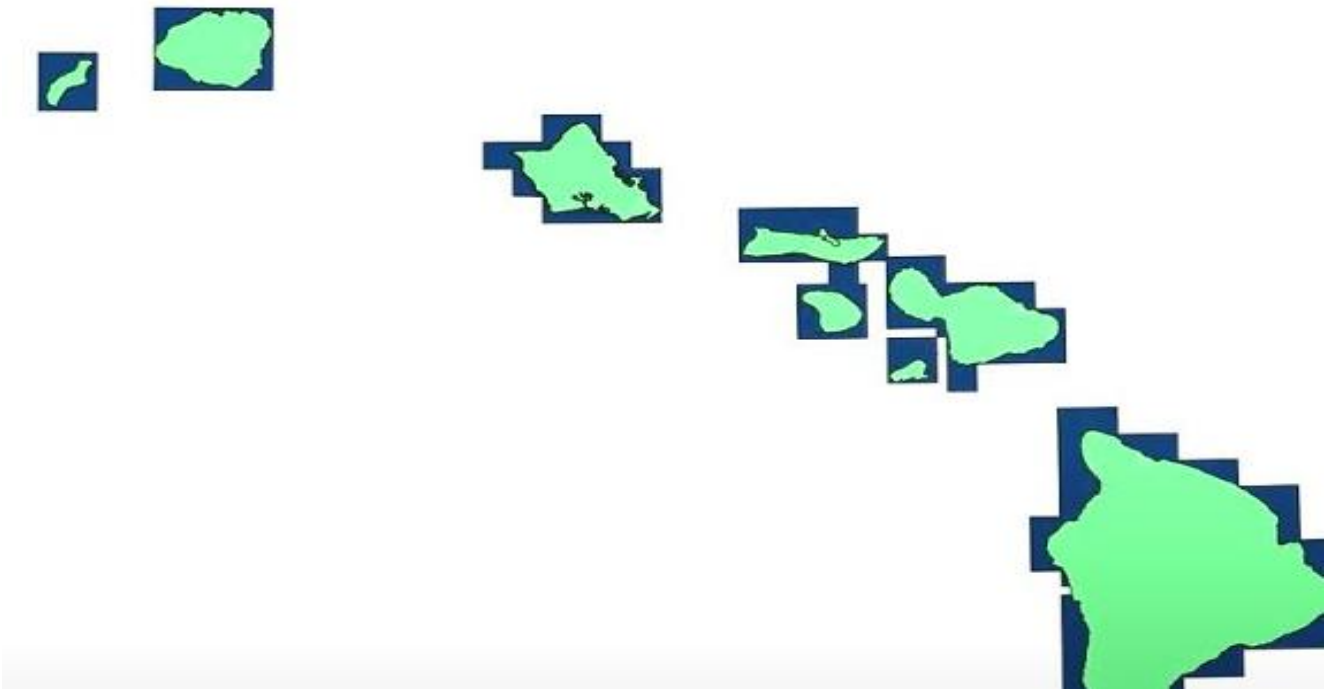
- Go to Project→New Print Composer.
- Leave the composer title field empty and click OK.
- Go to Layout →Add Map. Drag a rectangle while holding the left mouse button where you would like to insert the map.
- In the Item Properties tab and check the Controlled by atlas box.
- 10. Switch to the Atlas generation tab. Check the Generate an atlas box. Select the county10 as the Coverage layer. This will indicate that we want to create 1 map each for every polygon feature in the county10 layer.
- Go to Atlas ‣ Preview Atlas.



- Go to Atlas→Next Feature

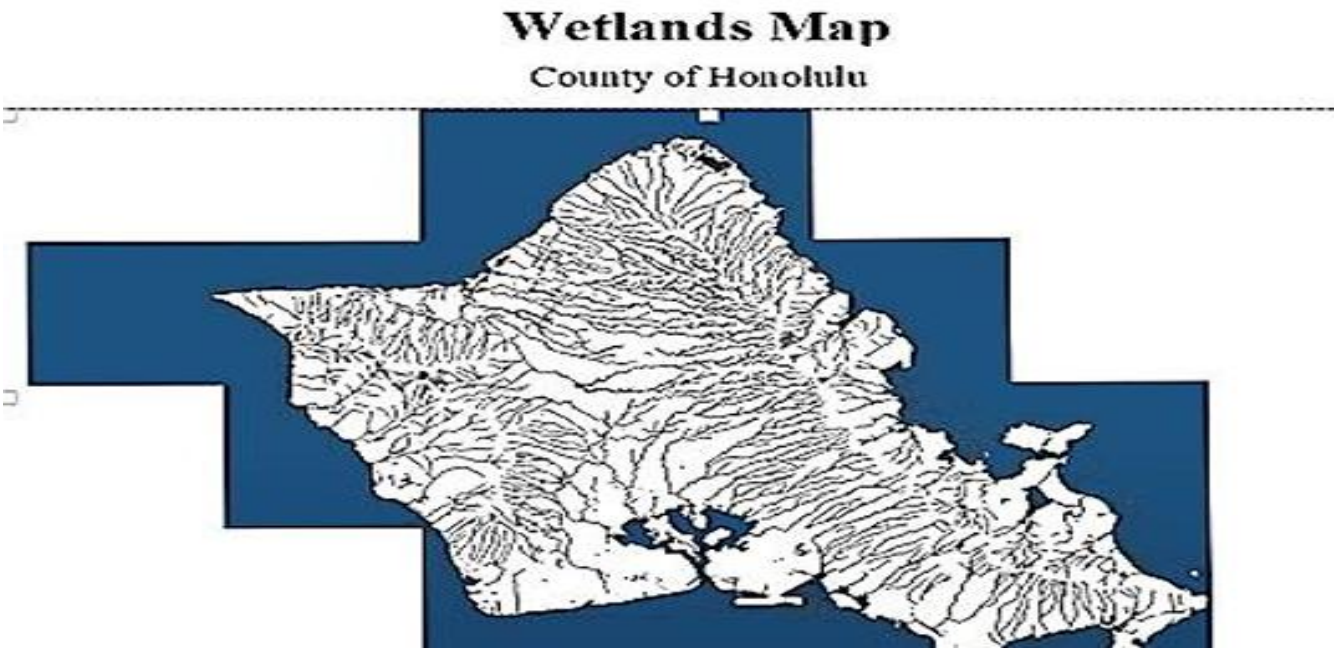


- Under the Item properties tab, click Insert an expression... button.
- Concat function is used to join multiple text items into a single text item. In this case we will join the value of the NAME10 attribute of the county10 layer with the Text County of. Add an expression like below & click OK.
- Add another label and enter Wetlands Map under the Main properties.
- Go to Atlas→Last Feature and verify that the map labels do work as intended.

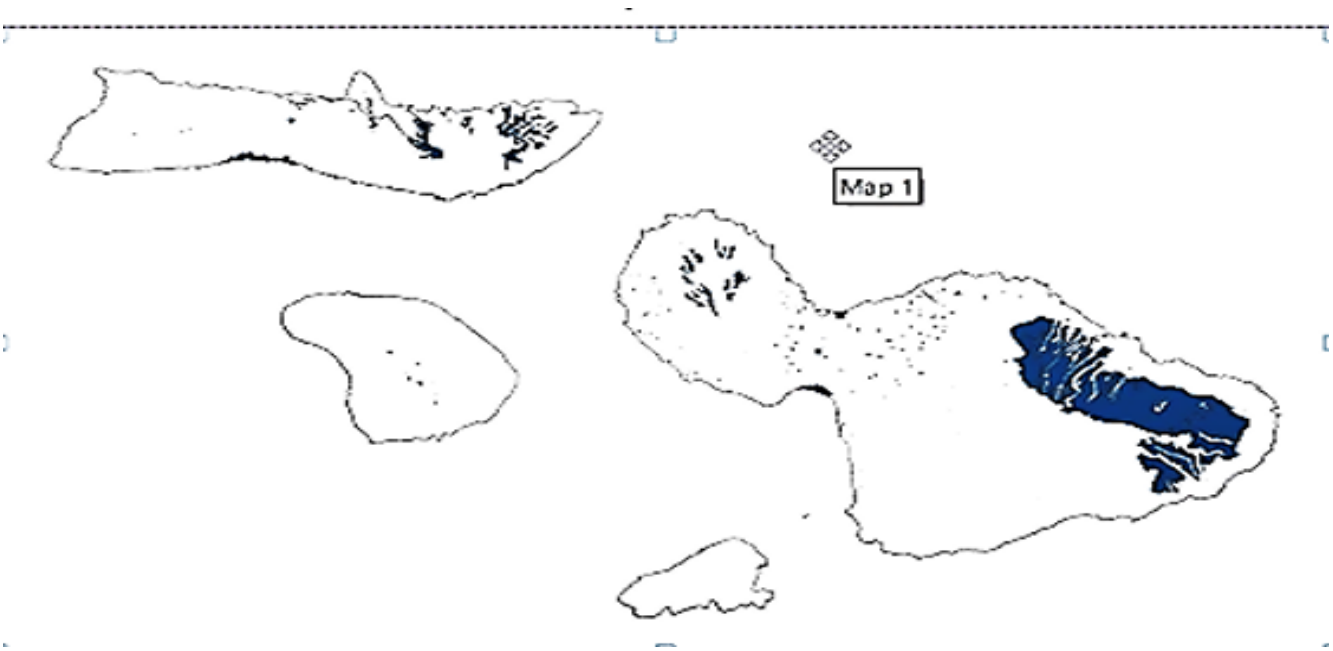


- Right-click the county10 layer and select Properties.
- In the Style tab, select the Inverted polygons renderer. This renderer styles the outside of the polygon - not inside. Select white as the fill color and click OK.

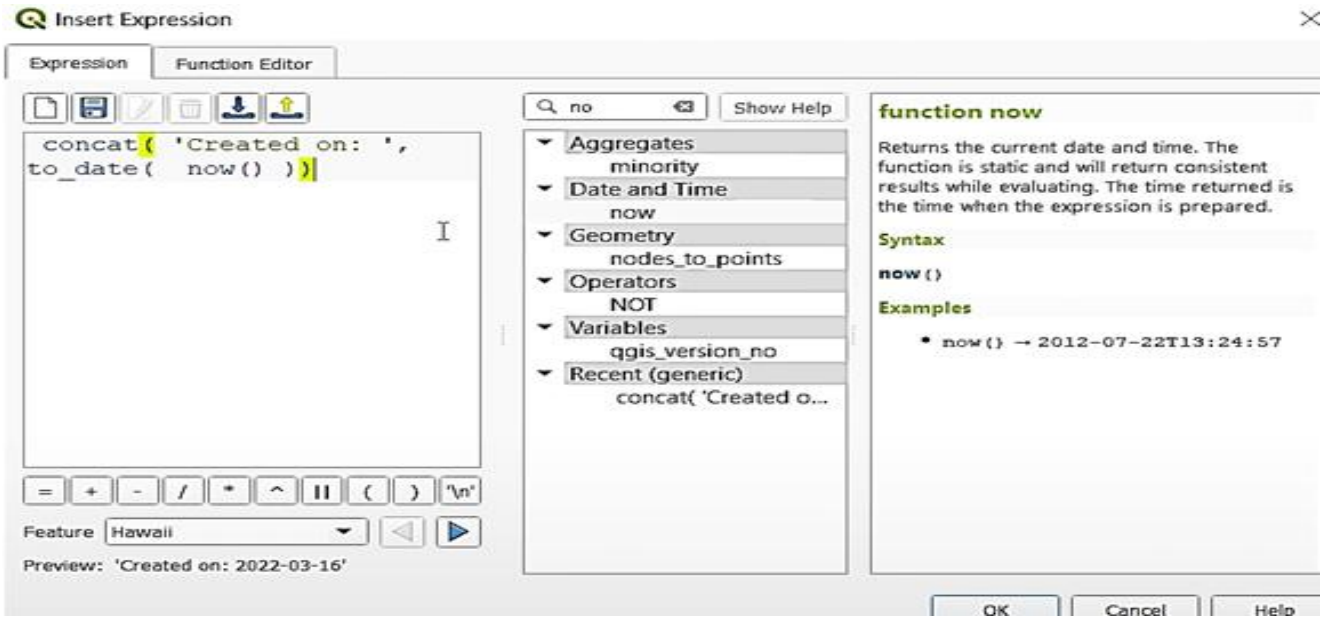
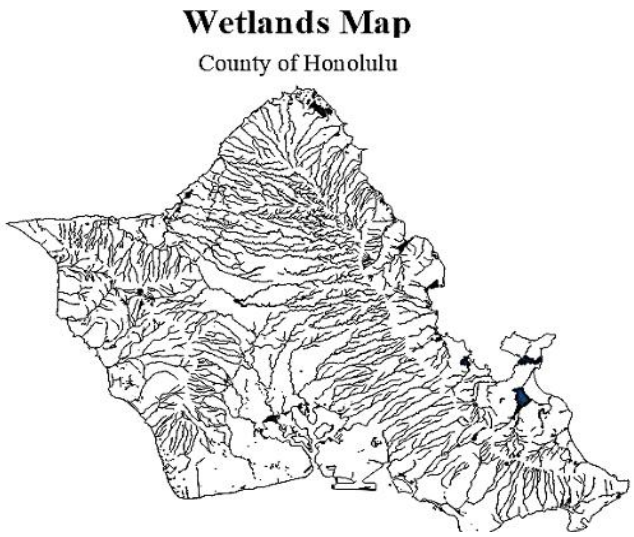
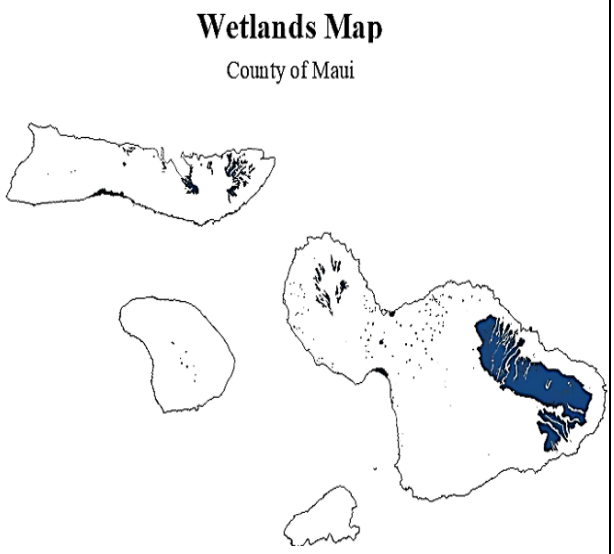
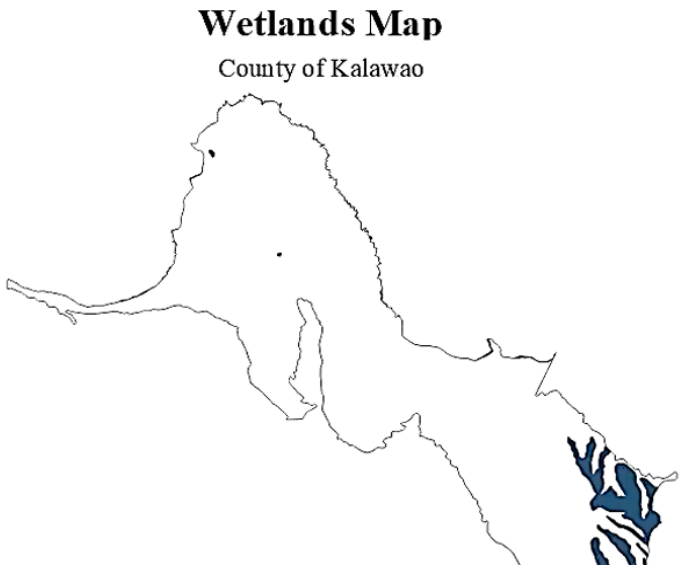
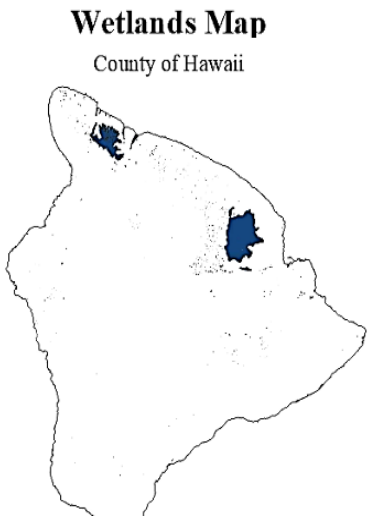
- Switch back to the main QGIS window and right-click the county10 layer and select Properties.



- In the Expression string builder, expand the Atlas group of functions. The $atlasfeatureid function will return the currently selected feature. We will construct an expression that will select only the currently selected Atlas feature. Enter the expression as: $id=$atlasfeatureid
- Back in the Print Composer window, click the Update preview button under Item properties tab to see the changes. Notice that now only the area covering the county boundary is shown



- Expand the Date and Time functions group and you will find the $now function. This holds the current system time. The function todate() will convert this to a date string. Enter the expression as below:



- Go to Atlas→Export Atlas as Images.
- Select a directory on your computer and click Choose.
- Here are the map images for reference.